

Czech Technical University in Prague
FJFI - Department of Physics
02ELMA - Electricity and Magnetism
Midterm 2

09.05.2025 / 14:30pm

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 90 minutes.
2. The maximum score for the test is 4 points and each question is worth 1 point.
3. Use of any class materials and calculators are not allowed.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. Write your answers in the provided space. If you need more space, additional sheets will be provided.

Question 1

A long, straight, solid cylindrical conductor of radius R carries a steady total current I in the $+\hat{z}$ direction. The volume current density varies with radial distance s from the axis as

$$\vec{J}(s) = \begin{cases} J_0 \frac{s}{R} \hat{z}, & 0 \leq s \leq R, \\ 0, & s > R. \end{cases}$$

- (a) Determine J_0 in terms of the total current I and the radius R . — *0.4 pts*
- (b) Using Ampère's law, find the magnetic field $\vec{B}(s)$ (magnitude and direction) for
 - (i) $s < R$, — *0.2 pts*
 - (ii) $s > R$. — *0.2 pts*
- (c) Sketch $B(s)$ versus s for $0 \leq s \leq 2R$. — *0.2 pts*

Solution:

Question 2

A long solenoid, of radius a , is driven by an alternating current, so that the magnetic field inside the solenoid is sinusoidal:

$$\vec{B}(t) = B_0 \sin(\omega t) \hat{z}$$

where B_0 is the amplitude of the magnetic field, ω is the angular frequency, and $\hat{z}$ is the unit vector in the direction of the solenoid's axis. A circular loop of radius $a/2$ and resistance R , is placed inside the solenoid, and is coaxial with it.

- (a) Find the current induced in the loop as a function of time. — *0.5 pts*
- (b) Find the power dissipated in the loop as a function of time. — *0.5 pts*

Solution:

Question 3

The magnetic field component of an monochromatic electromagnetic wave propagating in free space is given by plane waves as

$$\vec{B}(z, t) = B_0 \sin(kz - \omega t) \hat{y}.$$

- (a) What is the direction of propagation of the wave? — *0.1 pts*
- (b) Find the electric field component of the wave. — *0.2 pts*
- (c) Find the Poynting vector of the wave. — *0.2 pts*
- (d) How much power is carried by the wave through a circular area of radius a in the plane perpendicular to the direction of propagation? — *0.2 pts*
- (e) What is the direction of polarization of the wave? — *0.2 pts*
- (f) If the wave goes through a polarizer oriented at an angle $\pi/4$ with respect to the direction of polarization, what would be the amplitude of the transmitted wave? — *0.1 pts*

Given: $\vec{B}_0 = \frac{1}{c}(\hat{z} \times \vec{E}_0)$ where c is the speed of light in vacuum.

Solution:

Question 4

A circular loop of radius R carries a steady current I .

- (a) Write down the magnetic dipole moment $\vec{m}$ of the loop in terms of I and R . — 0.4 pts
- (b) Now, place the loop in a uniform external magnetic field $\vec{B}$ that points at an angle θ with respect to the normal vector of the loop. Write the torque $\vec{\tau}$ acting on the loop. — 0.6 pts

Solution: